

Compute requirements of detection and recognition

On the left side of the image in Fig. 2 you can see a trade-off between the accuracy versus the complexity. Some of the newer nn models allow you to get higher accuracy.

On the right side you can see the general number of operations per inferencing. Lot of these land around the 10-giga operations per inference, which is something we use to really understand the compute requirement that you need in order to do multiples of these while processing the video.

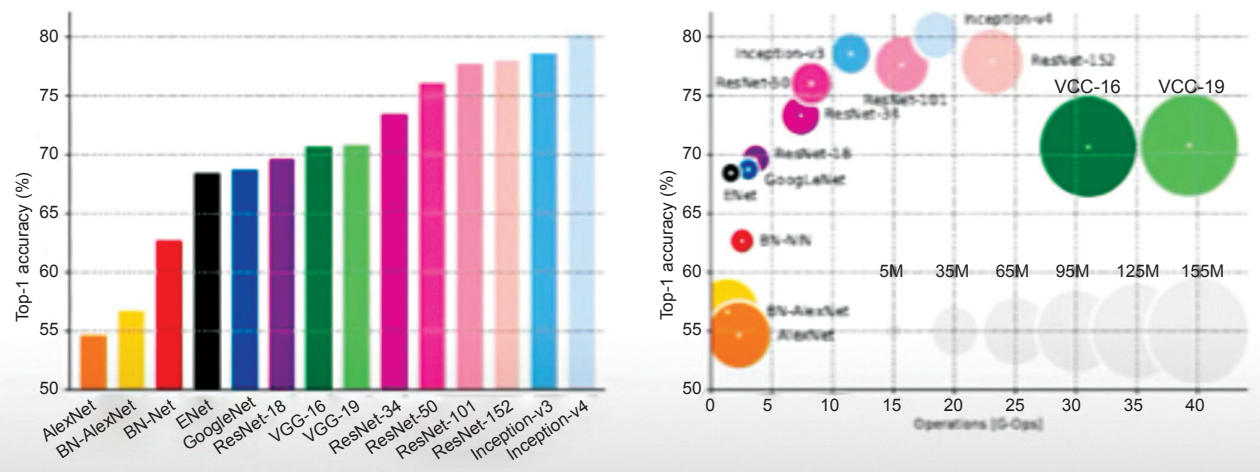


Fig. 2: Compute requirements of detection and recognition

image, and these types of specialised engines have the same characteristics, such as high energy efficiency.

So let us now look at some of the characteristics of these specialised engines.

Characteristics of specialised engines

All of them have a lot of processing as well as data movement involved. For instance, when there is a 12 megapixels being operated @ 60fps, it has a lot of pixels to process. In order to have high energy efficiency we try to avoid going to memory (DRAM). Many a time we try to do in-stream processing in order to save the energy that it takes to move the data around.

There are lots of registers that allow you to configure how the data is being processed but they do not have a lot of instructions to fetch and decode, such as the paradigm that you have in CPUs and GPUs. So, its programmability is limited, allowing you to gain a lot of energy efficiency.

Another characteristic is that they

rely on the classical approaches rather than newly developed deep learning based approaches. Thus, they also have somewhat simpler compute. In a lot of cases if a problem has a closed form solution, a classified configurable approach is the right approach.

Beyond classical approaches: Applications and compute requirements

Camera applications. These applications consist of two important steps: image understanding and image enhancement. For instance, Google Clip allows you to recognise a scene so you can use that to perform a capture decision.

The image enhancement based on the data or metadata that you have captured will let you take various approaches, such as getting a higher dynamic range or improving the resolution via multiframe processing. You can do a bokeh effect by using a single or multiple cameras. You can also do some adaptive scene

control in order to have the right look and feel for the images you capture.

Let us take a look at the detection side of things now.

Intelligent camera. We often try to detect the location of the objects with this camera as well the scenes themselves. A lot of times we try to get an object or a scene for which we can adaptively configure the data in an automatic fashion to get the best image quality.

If you combine ISP with a machine learning (ML) acceleration, it opens the door for you to do a plethora of things, such as scene detection as well as image quality enhancement using the captured metadata.

Pixel processing

Segmentation and single camera bokeh. You can apply the segmentation algorithm to a person and the background and then you can blur the background way more than the foreground using the segmentation map. It is important to note that at every pixel we need to make a classification for